

Programmable DC Power Supplies

Model	Description
Keithley-Instruments 2230-30-1	- Number of channels: 3 - Channel 1 and 2: (0-30)V/1.5A, Channel 3: (0-6)V/5A - Line regulation: Voltage ($< 0.01\% + 3\text{mV}$), Current ($< 0.1\% + 3\text{mA}$) - Ripple and noise: Voltage (7MHz) $< 1\text{mV rms}$, Current (20MHz) $< 5\text{mA rms}$ - USB, GPIB interface - Display type: VFD 
B&K Precision 1697 US\$ 355	- Number of outputs: 1 - Output voltage: (1-40)V, Output current: 5A - Line regulation: 50mV - Ripple and noise: 25mV, $< 10\text{mA rms}$ - RS232 interface - Display type: LCD 
Extech 382280 US\$ 649	- Number of outputs: 3 - Channel 1: (0-40)V/5A, Channel 2: 5V/2A, Channel 3: 3.3V/3A - Ripple and noise: 3mV, 3mA - Accuracy: $0.05\% \pm 9\text{mV}$, $0.02\% \pm 9\text{mA}$ - Display type: LCD 
Aplab PPD3003-S	- Number of outputs: 3 - Channel 1 and 2: (0-30)V/3A, Channel 3: (0-5)V/3A - Line regulation: Voltage ($< 0.01\% + 3\text{mV}$), Current ($< 0.1\% + 3\text{mA}$) - Ripple and noise: Voltage $< 1\text{mV rms}$, Current $< 5\text{mA rms}$ - GPIB interface - Display type: VFD 

Programmable AC Power Supply

Model	Description
B&K Precision 9801 US\$ 1995	- Number of outputs: 1 - Output voltage: (0-300)V AC, Output current: 3A - Crest factor: ≥ 4 - Line regulation: 0.1% maximum for a $\pm 10\%$ line change - Load regulation: $\leq 0.5\%$ FS (resistive load) - Response time: $< 100\mu\text{s}$ - Display type: VFD 

*Prices of the instruments are as listed on Mouser website

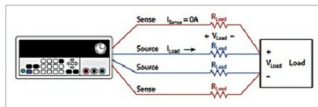


Fig. 2: Remote-sense capability

a power supply is largely due to error terms in the DAC, including quantisation error.

Read-back accuracy is sometimes called meter accuracy. It determines how close the internally-measured values are to the theoretical values of the output voltage (after setting accuracy is applied).

Read-back resolution is the smallest change in internally-measured output voltage or current that the instrument can discern. It is usually expressed as an absolute value, but may also be given as a percentage of full scale.

Load regulation. It is the measure of the ability of the output voltage or output current to remain constant during changes in the load.

Line regulation. It is a measure of the ability of the power supply to maintain its output voltage or output current while its AC line input volt-

age and frequency vary over the full allowable range.

Remote sensing. A programmable power supply is equipped with remote-sensing capability. Remote sensing is required in applications where load is located at some distance, typically more than 3m (10-feet), from the power supply output terminals. You can also use remote sensing if the measured voltage at the load input power terminals is significantly lower than the voltage measured at the power supply output terminals.

Difference in voltage is based on the amount of current and the load lead size and length. It uses a four-wire connection (Fig. 2) so that the voltage you set on the supply is the voltage you get at the device under test (DUT) in spite of voltage drops in cables that carry current between the power supply and DUT.

Analogue interface. DC programmable power supplies typically provide a standard and isolated analogue interface, through which a supply's DC output voltage, current and over-voltage protection can be set. These values are controlled by supplying a voltage signal, a current signal or by connecting a resistor to the analogue input. For example, you can use the analogue output of a PLC to control the output voltage of a power supply.

Residual AC. Output of these DC power supplies is not perfect DC. Some AC is to be expected on the output. For some applications excessive AC on the output can produce unexpected circuit behavior, so it helps to know the amplitude of the residual AC.

Spurious AC components on the output of a DC supply are called ripple and noise, or periodic and random deviation. These terms are often used interchangeably.

Transient response. Transient response specifications indicate how quickly the output settles to a stable DC value after a change in load or settings. Most power supplies have a significant capacitance in parallel